

Benchmark RADAR: Living Search Engine for Retrieval and Discovery of AI Benchmark Research

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Abstract

Benchmark researchers and model developers need to find relevant evaluations, locate their data and code, and understand the settings behind reported scores. We present Benchmark RADAR, a living search engine for retrieval and discovery of AI benchmark research. The system combines daily discovery of papers, repositories, datasets, and releases with a searchable benchmark catalog, model-report mentions, and score histories. It retains source identities and citations so readers can inspect candidate benchmarks and their evaluation evidence. At the 2026-09-07 cutoff, the catalog contains 1,283 source records across 4 sources and 12,916 numeric observations on 790 records. Of its 493 unscored records, 464 still link to task materials. We describe collection and retrieval, audit the full catalog, and examine reporting patterns and the limits of score comparisons. A contributor’s prior-art workflow illustrates how catalog queries and source inspection inform evaluation design. We release the web dashboard, feeds, downloadable evidence, offline query interfaces, and reproducible analysis.

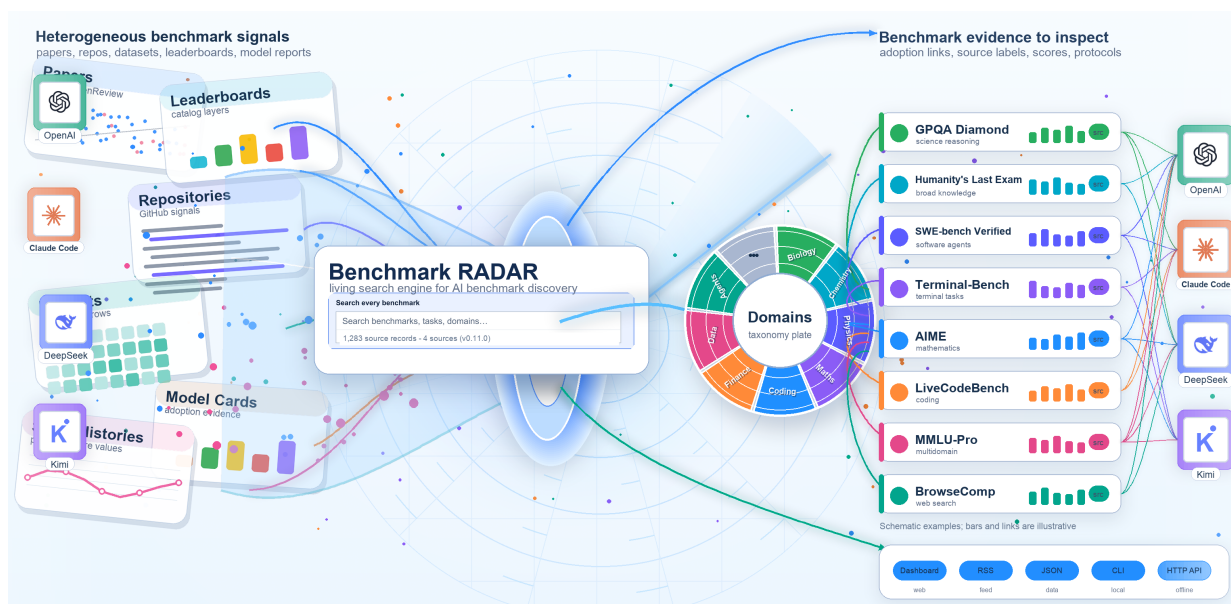


Figure 1: Benchmark RADAR connects discovery, retrieval, and source inspection. Named benchmarks, bars, and connections are schematic examples, not rankings or measured coverage. The catalog count uses the frozen release.

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1 Introduction

The Transformer architecture established attention-based sequence modeling as a foundation for modern large language models (LLMs) [44]. GPT-6 Astra and Claude Fable 5.1 are designed for coding, research, and tasks that span multiple tools [1, 30]. Comparing these models requires benchmarks that expose failures and distinguish capability gains from changes in prompts, data splits, or evaluation procedures [12, 32, 35, 39].

Evaluation also matters beyond general-purpose chat. In recommender systems, Transformer-style sequence models and LLM-based methods support user, item, and preference modeling, with efficient training needed at production scale [18, 37, 42]. Foundation models are also used to model biological, geoscientific, and physical processes [7, 38, 43, 46, 52]. In social and political analysis, language models support text annotation and simulated human samples, while network methods quantify balance and polarization dynamics [3, 9, 21, 22, 29, 40]. Benchmark scores inform claims of progress across these scientific, industrial, and decision-support settings.

Benchmarks differ in the abilities they test. Broad evaluations of knowledge and reasoning include MMLU [12], GPQA [35], and Humanity’s Last Exam [32]. Others isolate more specific capabilities: SWE-bench measures whether models can resolve real GitHub issues [17], LiveCodeBench tests code generation with date-based splits to reduce contamination from training data [16], and Terminal-Bench and Long-Horizon-Terminal-Bench test command-line agents on operational tasks [23, 25]. Domain-specific evaluations include FinanceBench for finance and economics [15], MATH for mathematics [13], SciBench for scientific problem solving [45], LAB-Bench for biology [20], and ChemBench for chemistry [26]. ESM-BENCH tests agents’ understanding of Earth system model physics and code [47]; ResearchClawBench evaluates end-to-end autonomous scientific research [49]; and ASI-Bench examines scientific exploration and execution with progressively less methodological guidance [53]. Their tasks and scoring rules specify which model behaviors count as successful performance.

Finding a benchmark, its dataset or repository, and reports of its use requires searching paper servers, code hosting sites, dataset hubs, vendor releases, blogs, and benchmark catalogs. Existing resources such as LLM Stats, OpenCompass, and Artificial Analysis provide leaderboards, evaluation platforms, and model-analysis views [4, 24, 31]. Benchmark RADAR centers on benchmark retrieval and discovery, combining these catalog records with daily discovery, mentions in model reports, and scores with documented evaluation settings and sources.

Benchmark RADAR gathers benchmark-related artifacts from public sources, groups observations with matching identifiers, and exposes a searchable index with source labels, dates, benchmark mentions, and scores. We present the search system, describe the collection and ranking methods needed to reproduce its outputs, and examine benchmark coverage, adoption in model reports, source concentration, and gaps in evaluation settings.¹

The system preserves one record per source benchmark, including entries without scores, dates, or citations. Reviewed identity links connect related records while retaining their separate measurements. We contribute this living search system, shared web and offline access to its evidence, and a reproducible full-catalog census. The census establishes the coverage behind the examples: 464 unscored records still link to task materials, and score comparisons require explicit scale, date, and protocol checks. Section 5 shows how a contributor used retrieval and source inspection to assemble prior art.

¹Source code: <https://github.com/ktwu01/benchmark-radar>.

2 Related Work and Scope

Benchmark catalogs and evaluations. LLM Stats publishes benchmark descriptions and reported model results [24]. OpenCompass provides an evaluation platform and a benchmark registry with artifact metadata [31]. Artificial Analysis publishes model evaluations and their methodology [4]. Benchmark Radar uses records from these sources alongside model-report evidence. Benchmark RADAR adds daily discovery, artifact histories, benchmark retrieval, and source inspection through shared web and offline access. These functions help readers investigate evaluations across the contributing catalogs; they do not change the evaluations themselves.

Documenting evaluation evidence. Model cards and datasheets motivate documenting evaluation conditions and dataset characteristics [8, 27]. Benchmark Radar retains that information when the collected sources provide it, with citations for follow-up review. Benchmarks such as GPQA [35], SWE-bench [17], and SciBench [45] define different tasks; placing their records in one catalog does not make their scores interchangeable. The census measures which evidence is available for inspection and which metadata still needs review.

3 System and Methods

3.1 System Overview and Daily Discovery

Benchmark Radar adapts BuilderPulse’s daily public-source collection approach [48]. Figure 2 separates the benchmark catalog from discovery history. Model reports include model cards, technical reports, system cards, and release posts; they contribute through the same record structure as benchmark registries.

Table 1 summarizes the four catalog sources and their primary uses in the frozen release.

Catalog source	Rows	Primary use
LLM Stats	687	Benchmark names and descriptions with source provenance.
OpenCompass Hub	461	Paper, repository, release, and dataset links with source-specific identity fields.
Artificial Analysis	25	Current commercial evaluation catalog entries.
Model reports	110	Benchmarks mentioned in model reports, including records without scores.
Total	1,283	All four sources, retaining each record’s source label.

Table 1: Catalog sources in the frozen v0.11.0 release. Rows count source-specific benchmark records, including records without scores.

3.2 Discovery Collection

Daily discovery observations describe mentions, releases, and updates found across public sources. An observation is one collected record; an artifact is a paper, repository, dataset, release, or page linked by exact identifiers. These objects differ from source-specific benchmark records. We retain their histories alongside the catalog without adding discovery observations to the benchmark total.

Each collection run searches a 48-hour window, records counts and errors by source, removes future-dated rows, and requires healthy core sources before publication. At the cutoff, arXiv [5], Hugging Face

Hub [14], and GitHub Search [10] were the core sources. Additional routes cover scholarly indexes, dataset hosts, repository releases, and institutional feeds. Appendix C records their cutoff status.

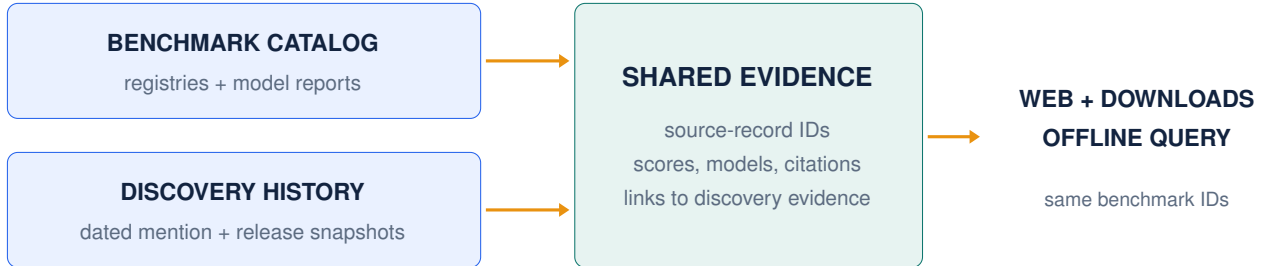


Figure 2: Two input paths support publication: dated discovery snapshots preserve mention and release evidence, while benchmark registries and model reports populate the shared catalog. Clients use the catalog’s stable source-record IDs.

Table 2 describes the interfaces. Search and exports retain the full catalog; display filters affect only the selected view.

Interface	Reader question	Evidence and scope
Today	What appeared or changed?	Daily discovery observations, source health, and links.
Search	Which benchmarks match this task?	The complete catalog, including unscored records.
Leaderboard	Which evaluations combine lower scores with broader measured use?	Scored source records; distinct scored models or cited documents supply the selected count.
Saturation	What scores and settings were reported?	Benchmark browsing and complete score histories; search reaches every source and year.
CLI and HTTP	Can I inspect the same evidence locally?	One query service and the same benchmark IDs and response format.

Table 2: Reader questions and evidence scope. A search result is a candidate for inspection, not a suitability judgment.

3.3 Preserve Records Before Comparing Measurements

A source record is one benchmark entry from one contributing source. We normalize names, identifiers, artifact links, score observations, model identities, and cited documents into common fields. A record remains in the catalog if a score, date, or citation is absent. Reviewed identity links connect related records while preserving their separate observations and counts. A similar name alone does not authorize a merge.

Daily discovery contributes a different kind of evidence: collected mentions, releases, and updates. Exact identifiers such as DOIs, arXiv IDs, and repository URLs link those observations to artifacts. Discovery observations do not increase the benchmark catalog total. Appendix B records detailed collection settings and display filters; Appendix C records source health at the cutoff.

3.4 Retrieve Candidates with Their Evidence

The same benchmark IDs reach web search, detail pages, dataset exports, and offline clients. Lexical search uses BM25F, a field-weighted word-matching score [36], with bounded boosts for name and phrase matches. Each result exposes matched and missing query words, the fields they occur in, and the score components. Source membership does not change the ranking. A shared query service supplies the CLI and HTTP interfaces with the same response format and local data provenance.

Candidate retrieval precedes suitability judgment. An analyst or agent can try focused query variants, inspect a record’s tasks and score settings, and follow its citations. The interface preserves the evidence needed for that review. This paper evaluates catalog and measurement coverage; the contributor case illustrates usage without measuring retrieval accuracy or time saved.

3.5 Audit the Full Population

The census starts with every record in the rebuilt catalog index and reads its detail file. It applies no date, score, or interface filter. We count finite numeric score observations once by observation ID. Within each benchmark record, we count scored models by source model ID, preserving separately evaluated configurations, and cited documents by document ID. Repeated observations do not create additional models or documents. The global model registry uses its recorded identity links; per-benchmark model counts are not summed into a global total.

Eligibility depends on the measurement a calculation needs. A declared percentage unit, known score direction, and numeric values within 0–100 permit a percentage-scale summary. Rescaling a displayed value or reading an aggregator’s declared maximum does not establish that unit. Matching scales also do not establish matching test versions, prompts, tools, attempts, or evaluators.

Date coverage counts valid recorded benchmark release dates. Score entries retain their own date basis, including model announcements and document publication. We do not substitute those dates for an evaluation date. Appendix E provides the software revision, input hashes, census script, and validation procedure.

4 Results

Figure 3 shows how a retrieved benchmark opens into reported model scores. The full-population results below retain every source record; Appendix A provides the complete linked census.

4.1 Unscored Records Still Provide Routes to Prior Work

The rebuilt catalog contains 1,283 source records across 4 sources. We found 12,916 numeric score observations on 790 records, with 493 records lacking numeric scores (Table 3). These are source-record counts: two sources can describe a related benchmark, and we retain both records without claiming they represent distinct underlying tests.

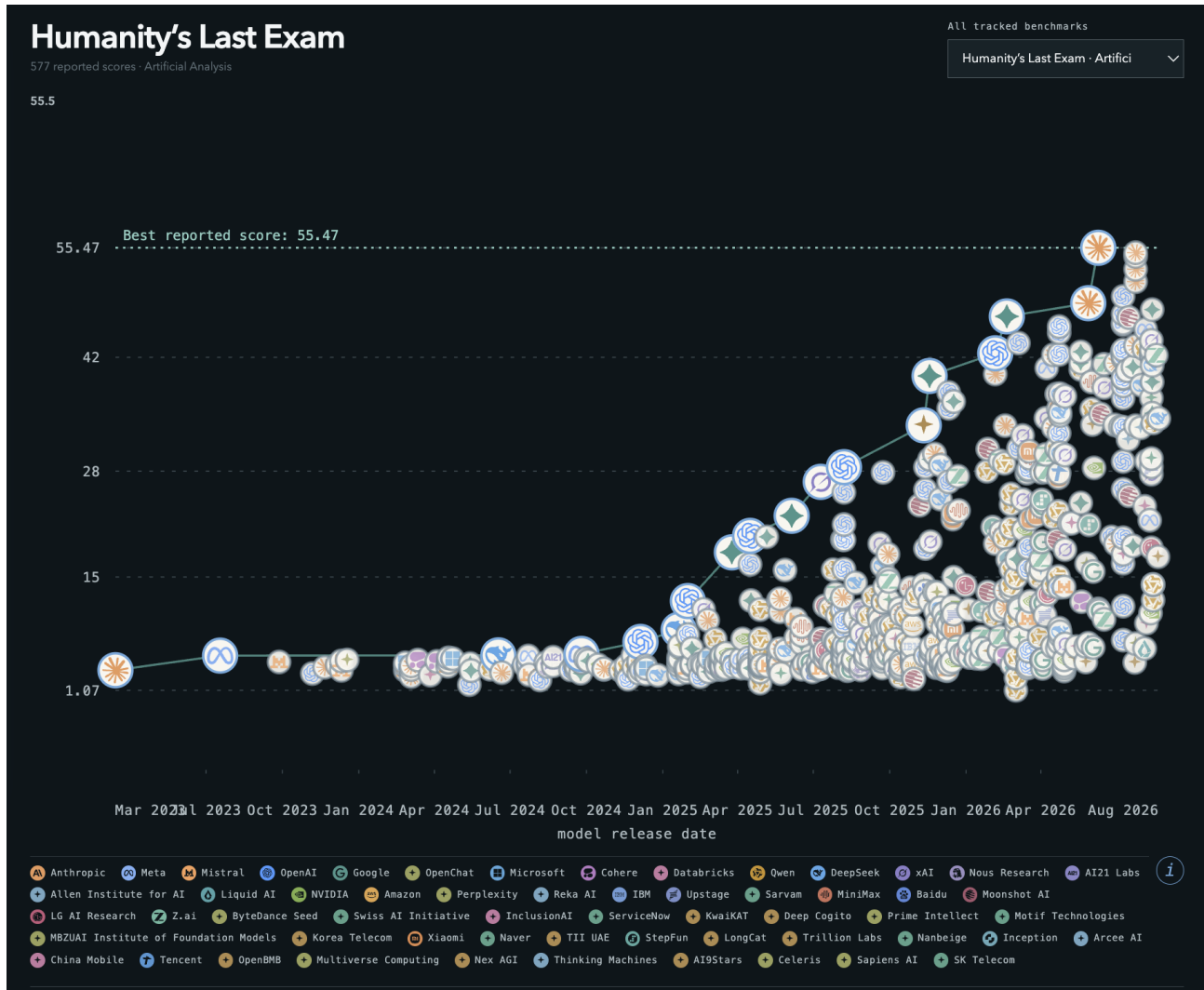


Figure 3: Inspecting Humanity’s Last Exam in Benchmark RADAR. This Artificial Analysis record contains 577 reported scores; the highest displayed score is 55.47. Marks identify model organizations, and the line connects successive best scores ordered by model release date. These are not evaluation dates: older models can be tested after a benchmark’s release. The snapshot illustrates score inspection, without establishing matched evaluation settings or a validated percentage ceiling. Click the image to open the interactive view.

Source	Records	Scored	Unscored	Numeric scores
LLM Stats	687	679	8	5,544
OpenCompass Hub	461	0	461	0
Artificial Analysis	25	25	0	7,050
Model reports	110	86	24	322
Total	1,283	790	493	12,916

Table 3: Score coverage across the full catalog: 790 records have numeric observations and 493 do not. The zero in the OpenCompass score column describes this snapshot, not whether those benchmarks have ever been evaluated. Scores count observations, not distinct models.

Of the 493 unscored records, 464 have at least one paper, repository, or dataset link. Across the full catalog, 475 records have a paper link, 506 have a repository link, and 293 have a dataset link; the categories overlap. For example, the OpenCompass Hub record for A-Bench links a paper, code repository, and dataset despite having no archived score. These links give researchers routes to inspect tasks and baselines before deciding whether an evaluation fits their work. Link presence does not establish current availability or permission to use an artifact.

4.2 Document Counts Do Not Measure Model Coverage

The common document registry contains 1,208 distinct cited documents, attached to 1,278 of the 1,283 benchmark records. Five records have no cited documents. The shared model registry contains 868 model identities. Within the catalog, all 790 scored records have sufficient model IDs to count distinct models; unscored records have unknown scored-model coverage.

Table 4 illustrates why these counts cannot substitute for one another. The Artificial Analysis record for GPQA Diamond contains scores for 586 models and cites one registry page. The Model reports record cites 27 documents and contains 21 numeric observations for 19 models. Both describe useful evidence, but adding their counts would require reviewed model and evaluation identities. Neither row alone measures adoption across the field.

Benchmark record	Source	Models	Documents	Scores
GPQA Diamond	Artificial Analysis	586	1	586
GPQA Diamond	Model reports	19	27	21
Humanity’s Last Exam	Artificial Analysis	577	1	577
SciCode	Artificial Analysis	577	1	577
CritPt	Artificial Analysis	492	1	492

Table 4: One cited page can document hundreds of scored models. Models count distinct source model IDs within a record; documents count distinct citation IDs; scores count numeric observations. These selected examples illustrate the units. The census retains all source records.

Table 4 also records 577 scored models for Humanity’s Last Exam and SciCode, and 492 for CritPt, each under one Artificial Analysis citation. Reading that citation count as model coverage would hide the measured breadth of those records.

4.3 Numeric Scores Often Lack a Verified Common Scale

Only 82 of the 790 scored records declare a percentage unit with a known direction and values within 0–100. The remaining 708 scored records use other or unverified scales. We retain their numeric observations, which do not support a shared percentage-headroom calculation. Table 5 accounts for the entire catalog before any such comparison.

Measurement state	Records	Interpretation
Numeric scores, declared percentage scale	82	Supports scale-specific summaries; protocols still need checking.
Numeric scores, other or unverified scale	708	Retain scores; do not assume a 100-point ceiling.
No numeric score	493	Score headroom remains unknown.
Full population	1,283	Every source record remains in the census.

Table 5: Only 82 of 790 scored records meet the percentage-scale rule in Section 3.5. All 1,283 records remain accounted for. A declared scale alone does not establish a common evaluation protocol.

The catalog records benchmark release dates for 615 records, leaving 668 without a release date. Across score observations, 12,594 entries carry model-announcement dates and 322 carry document-publication dates. Neither date basis directly establishes an evaluation date. This archive therefore cannot support a catalog-wide claim about score stagnation from a gap between model-report mentions and score-entry dates. Researchers must first establish dates and comparable evaluation settings for the relevant source records.

A small gap to a metric ceiling can describe a reported setup. It does not show that a benchmark has exhausted its ability to distinguish models across versions and settings. Catalog-wide headroom or recency estimates require the eligible measurement coverage alongside the statistic.

Figure 4 puts these coverage distinctions into a browsing view: readers can locate a benchmark by date, reported score, and the number of scored models, then inspect its source record.

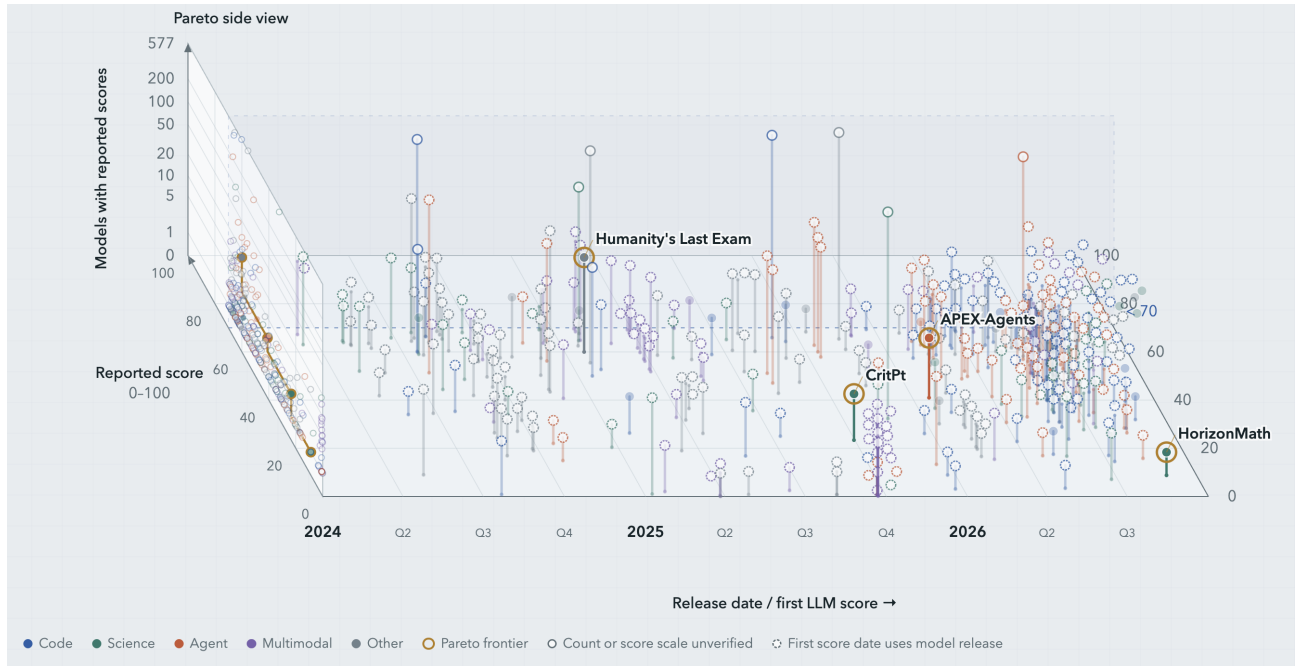


Figure 4: Benchmark Frontier on the Leaderboard page. The view combines reported scores, scored-model counts, and benchmark release dates or first-score date proxies. Gold rings mark the interface’s Pareto candidates; hollow marks flag unverified scales or counts, and dotted outlines flag model-release date proxies. This filtered, scored view supports candidate inspection; it does not replace the full-catalog census or establish equivalent evaluation settings. Click the image to explore the view.

4.4 Discovery Coverage Alongside the Catalog

The daily discovery collection contains 11,068 observations and 6,546 artifacts linked by exact identifiers across 46 snapshots. Four snapshots are simulated historical backfills; their dates describe reconstructed collection windows. These discovery units remain separate from benchmark records.

Five discovery source labels account for 9,743 of 11,068 observations (88.0%). Figure 5 includes the remaining labels in one aggregate bar. Source caps and collection failures can affect this mix, so it does not estimate the distribution of benchmark research. Fifty-nine of 6,546 artifacts have observations from multiple sources; missing or inconsistent identifiers can leave further relationships unresolved.

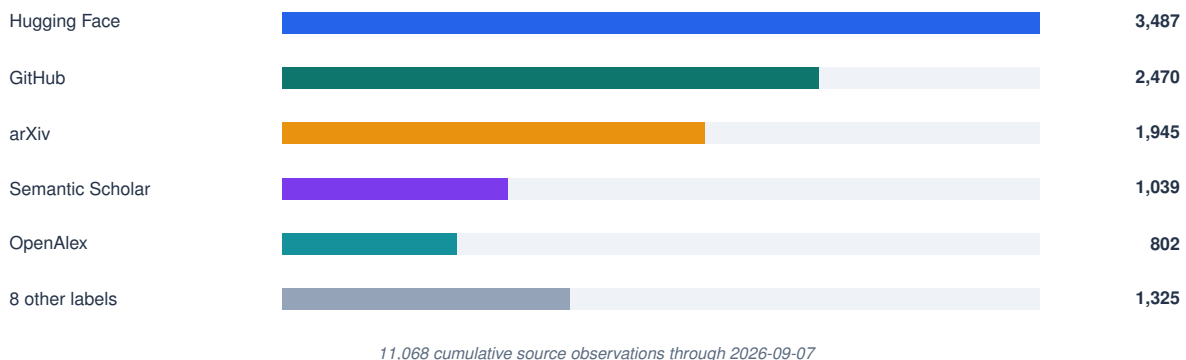


Figure 5: Discovery observations by source through 2026-09-07. Repeat sightings count as observations; these bars do not count benchmark records.

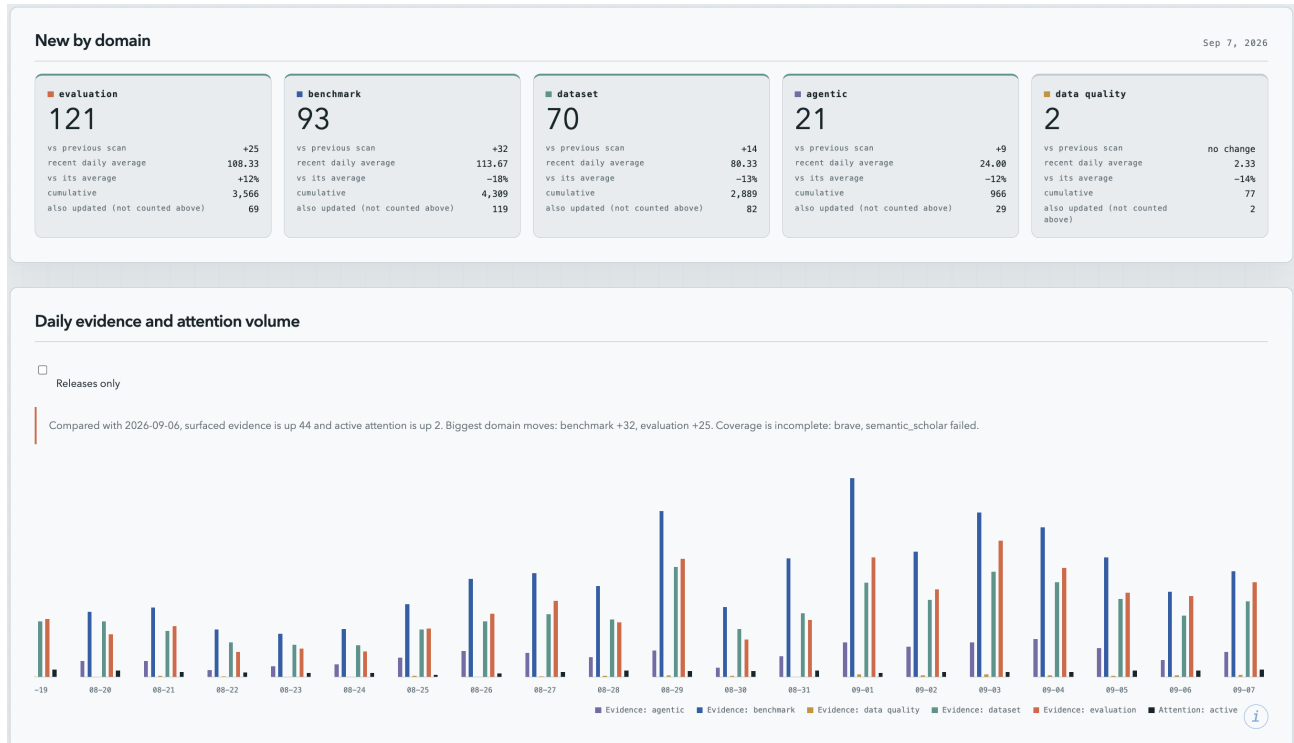


Figure 6: Discovery trends through September 7, 2026. Category cards show newly surfaced releases and track updates separately; the bars show daily evidence and attention volume. Category tags overlap, so these counts cannot be added into a total of unique benchmarks. The visible coverage notice reports failed sources. This is a view of collected activity, not research quality or benchmark performance. Click the image to open Trends.

Figure 6 complements source coverage with the daily reading view. Its category cards separate new releases from updates, while the chart exposes changes in surfaced evidence and attention alongside collection failures.

4.5 Benchmark Mentions in Model Reports

The catalog includes 37 model reports from 12 organizations. Within that document collection, 16 benchmark records are mentioned by at least six organizations (Table 6). GPQA Diamond appears in 27 of the 37 documents from 11 organizations. The frequently mentioned records span scientific reasoning, coding, agents, mathematics, and multilingual or multimodal tasks. This describes reporting choices in the collected documents; it does not measure field-wide adoption or establish that different versions of a named benchmark are equivalent.

Benchmark record	Documents	Organizations
GPQA Diamond	27	11
Humanity’s Last Exam	20	9
Terminal-Bench	19	9
SWE-bench Verified	18	8
AIME	17	8
LiveCodeBench	17	8
MMLU-Pro	12	7
SWE-bench Pro	10	7
IFEval	9	7
MMLU	9	7
BrowseComp	12	6
MATH-500	10	6
MMMU	10	6
HumanEval	9	6
GSM8K	7	6
MMMLU	7	6

Table 6: Every model-report benchmark record mentioned by at least six organizations. Counts deduplicate document IDs and organization names within each record. This scoped analysis starts from the full catalog; its report evidence covers 105 of the 110 Model reports records. The accompanying audit lists each document and source URL.

4.6 Near-Ceiling Reported Scores

Across the full catalog, 82 records declare a percentage scale, have higher-is-better direction, and contain numeric values within 0–100. Of these, 8 have a highest archived value within five points of 100. All eight qualifying records come from Model reports; 708 scored records on other or unverified scales and 493 unscored records remain in the census with unknown upper-ceiling headroom. Table 7 gives the model, metric, and recorded settings behind each example.

Headroom is 100 minus the reported value for that setup. It describes the remaining range of its metric, without establishing benchmark saturation or comparing models across settings. The values are maxima within the archived source records, not necessarily the best results in the cited documents or the field. The examples cite the GLM-5.2, GLM-5, Qwen3, Kimi K3, Claude Opus 5, and Gemini 3.1 Pro reports [2, 11, 28, 33, 50, 51].

Benchmark	Model and source	Score	Gap	Metric and recorded settings
AIME	GLM-5.2 [source]	99.2	0.8	pass@1; AIME 2026, temp 1.0, top_p 0.95, max gen 163840 tokens, GPT-5.5 (medium) judge
Arena-Hard	Qwen3-235B-A22B (Thinking) [source]	95.6	4.4	win_rate; thinking mode
DeepSearchQA	Kimi K3 [source]	95	5	f1; F1, reasoning=max
HMMT	GLM-5 [source]	96.9	3.1	accuracy; HMMT Nov. 2025, temp 1.0, top_p 0.95, GPT-5.2 (medium) judge
MATH-500	Qwen3-235B-A22B (Thinking) [source]	98	2	accuracy; thinking mode
MathVision	Kimi K3 [source]	97.8	2.2	accuracy; reasoning=max, with Python
SWE-bench Verified	Claude Opus 5 [source]	96	4	resolved; adaptive thinking, max effort, averaged over 5 trials
tau2-bench	Gemini 3.1 Pro [source]	99.3	0.7	accuracy; Thinking (High), Telecom

Table 7: Highest archived scores for the 8 near-ceiling source records. All qualifying records in the full-catalog scale audit are from Model reports. Scores use 0–100 scales; gaps are percentage points below 100. The metrics include accuracy, pass@1, F1, resolved-task rate, and win rate. Each row links to the reporting source registered in the frozen catalog.

The test version and settings determine what each small gap means. AIME 2026 and AIME 2024 use different problem sets; the 99.2 result above and the archive’s older AIME 2024 result of 79.8 describe different tests. Likewise, the tau2-bench value above concerns Telecom, and the MathVision value includes Python access. Benchmark Radar preserves these details so readers can select the relevant test and inspect the score in context. Comparisons require checking the source documents for matching versions, reasoning budgets, tools, attempts, and evaluators [8, 27].

4.7 Later Mentions Identify Gaps in Archived Scores

Six benchmark records have a later model-report mention at least 180 days after the document supplying their latest archived numeric score (Table 8). The gaps range from 343 to 775 days. These records give a reader newer reports to inspect for possible score extraction. Both endpoints are recorded document-publication dates, not evaluation dates; revised reports can also acquire mentions after first publication. The intervals describe the archive’s document coverage and cannot establish a period without benchmark progress.

Benchmark	Score document	Mention document	Gap (days)
GSM8K	2024-03-08	2026-04-22	775
HumanEval	2024-09-18	2026-04-22	581
DROP	2025-01-22	2026-04-22	455
MMLU	2025-01-22	2026-04-22	455
MBPP	2024-09-18	2025-11-20	428
MATH-500	2025-05-14	2026-04-22	343

Table 8: All six records meeting the 180-day archive-gap rule. Each date links to its source document. Dates and scores remain attached to the same source record; no similarly named aggregator record supplies a replacement score date.

5 Worked Example: Checking Prior Art

Before designing a new evaluation, a contributor surveyed August work on credit assignment in agentic training, with small Qwen-series models as a requirement for reproducible baselines. A coding agent installed the Benchmark RADAR client and its public Skill, downloaded the corpus, and searched locally. It inspected the recorded paper, repository, and dataset links, tried additional web searches for work described in different terms, and read the source evidence before assembling the related-work table in Figure 7.

The contributor used the table to assess whether the proposed evaluation duplicated existing work. The workflow separates candidate retrieval from comparison: Benchmark RADAR retrieves candidates and exposes their evidence; the researcher or agent judges their relevance and compares the designs. The session has no measured time or accuracy baseline and combines local retrieval with web search. Appendix D retains the session screenshots and an earlier manually assembled comparison.

#	Work	Published	Qwen base model(s)	Credit-assignment focus	Agentic benchmarks used	Link
1	SRPO: Self-Reflective Policy Optimization for Long-Horizon Reasoning	2026-08-25	Qwen3-8B (also Qwen3-1.7B / 32B scaling)	Reflection-conditioned dense token-level signals convert sparse terminal reward into training signal	AIME'24, WebShop, ALFWorld, SWE-Bench-Lite	https://arxiv.org/abs/2608.23493 · https://github.com/Galleons2029/SRPO
2	ContextPilot: Teaching Agents for Proactive Context Management via Fine-grained RL	2026-08-28	Qwen3-8B, Qwen3-14B	Coarse-grained → action-level credit assignment over branched trajectories during agentic RL	InfBench (InfiniteBench), NovelQA, LongMemEval, BrowseComp+	https://arxiv.org/abs/2608.28476 · https://github.com/Tencent/ContextPilot
3	SkillGate: Training In-Policy Skill Selection in Long-Horizon Agents	2026-08-21	Qwen3.5-9B (SFT checkpoint, RL init)	"Selector credit starvation": partitions token support into two credit channels (outcome vs. action-local advantage)	5 agentic benchmarks incl. SkillsBench, SWE (16-candidate skill slate)	https://arxiv.org/abs/2608.18852
4	CIPO: Contextual Information Policy Optimization for Search Agents	2026-08-06	Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct	Dense turn-level credit (EALR) to evidence-using reasoning actions, combined with outcome reward	HotpotQA, 2WikiMultiHopQA, MuSiQue, Bamboogle + 3 OOD	https://arxiv.org/abs/2608.06128
5	MoRSE: Task-Oriented Multi-Agent System with Mixture of Role-Subtask Experts	2026-08-10	Qwen3-4B-Instruct (one of three backbones; also Llama-3.1-8B, Gemma-4-31B)	Hierarchical GRPO with two-layer credit assignment (isolates expert vs. routing quality)	Code-generation benchmarks (multi-agent), held-out task domains	https://arxiv.org/abs/2608.09251

Figure 7: Summary table of recent work on credit assignment in agentic training, assembled during the session.

6 Limitations and Future Work

The census describes this catalog at its recorded cutoff. Collection limits, failed requests, missing identifiers, and different snapshot dates affect its coverage. Source records are not a count of distinct underlying tests. Artifact links provide routes for inspection but do not verify present availability or licensing. Broader coverage requires additional source collection and review of the evidence already present.

The study does not measure retrieval precision, task suitability, or time saved. The worked example combines local queries with web search and has no controlled baseline. Lexical matching can miss paraphrases and renamed tasks. Evaluating semantic retrieval [34] will require reviewed relevance judgments across queries and candidate records, including less familiar benchmark families.

The scale and date audit identifies metadata gaps. The restored score and mention examples describe specific archived evidence, without establishing benchmark saturation or score stagnation. Establishing those outcomes requires source review for comparable test versions and settings, and dates tied to actual score reporting or evaluation. Matching archive labels cannot verify independent repeated runs.

Task-capability classification also remains unvalidated in this rebuild. The deterministic null extractor used in CI assigns no capability levels to its 5,863 discovery-derived tracks. Completing and evaluating those labels is a separate task from maintaining the benchmark catalog.

7 Conclusion

Benchmark RADAR brings daily benchmark discovery, catalog search, model-report mentions, and score histories into one living search engine. Readers can find candidate evaluations, locate task materials, inspect reporting choices, and follow the settings and citations behind a score. Shared benchmark IDs connect the web dashboard, downloadable catalog, and offline query clients.

The frozen release makes 1,283 source records searchable and preserves 464 unscored records with artifact links. Its full-catalog census and scoped examples distinguish benchmark records, scored models, cited documents, and numeric observations. Further work should evaluate retrieval with reviewed relevance judgments, resolve missing artifact identities, and recover the protocol and date evidence needed for score comparisons.

Author Contributions

Koutian Wu led the work and manuscript preparation, built the initial collection pipeline, and implemented data aggregation across sources. Junjie Zhou prepared the score-archive audit and contributed report revisions. Ergan Shang prepared figures and contributed copyediting. Jiayu Wang contributed the worked use case and its supporting evidence. Pengqian Han contributed copyediting.

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Appendix

A Full-Catalog Census

The score view in Figure 3 selects one benchmark. This census retains all 1,283 source records, including those without scores, so that the selected example does not obscure the catalog’s discovery coverage.

1,283 records. 790 with scores. 493 without.

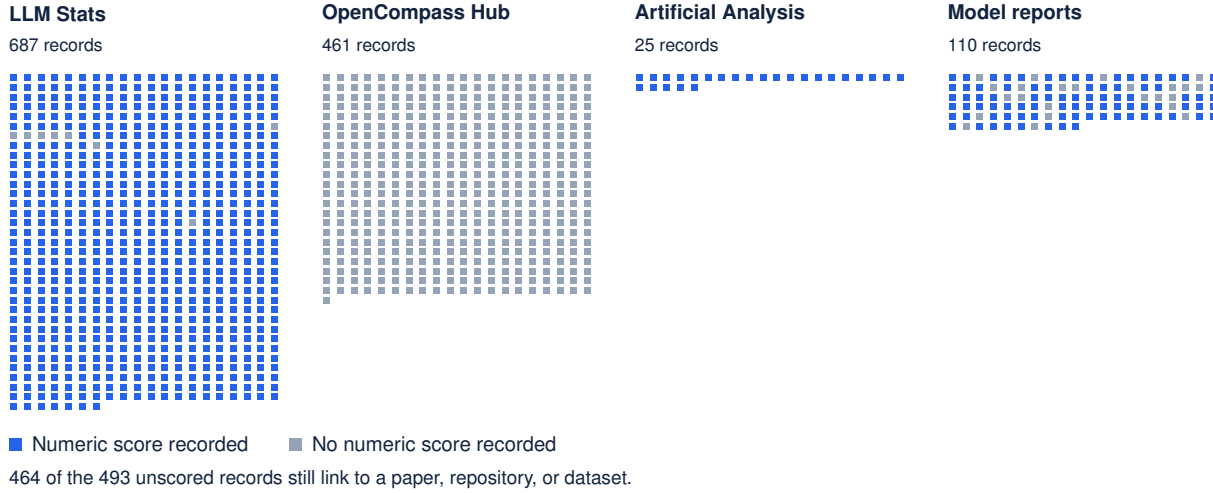


Figure 8: Discovery extends beyond scored records: 464 unscored entries still provide artifact links. Each mark represents one source record and links to its detail page. All 1,283 records appear, without a date or score filter.

B Collection Settings and Display Filters

The 2026-09-07 snapshot records 1,003 fetched rows and 954 candidates after duplicate removal. Of these, 366 qualified for publication and 138 met the recommendation threshold. These are discovery-run counts, separate from the benchmark catalog census. Published evidence retains source URLs, retrieval times, parser versions, identifiers, and payload checksums. Raw responses and credentials remain outside the public artifact.

Benchmark Frontier, on the Leaderboard tab, requires a numeric score and excludes known pre-2024 benchmarks. It retains scored records with unknown dates or counts through labelled marks, and keeps unverified scales outside Pareto calculations. Its initial score cutoff is 70. The linked score ranking retains its date and numeric-score requirements independently of that cutoff. General catalog search and exports retain the full population. The census in this paper applies none of these display filters.

A separate priority score for daily recommendations combines relevance (35%), evidence (20%), recency (20%), and adoption (25%) on a 0–100 scale. These weights describe reading priority for discovery observations. An agent assessing benchmark suitability can try focused query variants, inspect record details, and read the cited sources; that judgment occurs after candidate retrieval.

C Source Inventory and Health

Table 9 lists source connectors that completed successfully at the cutoff. Row counts are measured before duplicate removal and ranking. A connector can be healthy with zero eligible rows if its request succeeds and the response can be parsed. Core sources cover papers, shared model or dataset artifacts, and code; all must be healthy before publication. Failures in optional sources are reported but do not block publication.

Connector	Role	Cutoff status	Core
arXiv	Primary papers via selected AI, language, vision, and software categories	Healthy; 46 rows	Yes
Hugging Face Hub	Datasets and Spaces connected to benchmark artifacts	Healthy; 101 rows	Yes
GitHub Search	Code repositories and benchmark artifacts	Healthy; 300 rows; at cap	Yes
GitHub Organizations	Reviewed organization repositories	Healthy; 6 rows	No
Hugging Face Papers	Community-surfaced papers	Healthy; 26 rows	No
Kaggle Datasets	Public benchmark datasets	Healthy; 30 rows	No
Zenodo	DOI-bearing research artifacts	Healthy; 65 rows	No
Crossref	DOI metadata	Healthy; 180 rows	No
OpenReview	Conference submissions	Healthy; no eligible rows	No
GitHub Releases	First-party releases	Healthy; 1 row	No
OpenAlex	Scholarly discovery metadata	Healthy; 248 rows	No

Table 9: Healthy direct discovery connectors at the current cutoff.

Table 10 lists research and engineering feeds operated by the institutions being monitored. They can surface benchmark papers, datasets, model announcements, and engineering posts. The feed collector returned no records at the cutoff. Semantic Scholar returned a malformed payload, and Brave Search lacked an API key. These optional-route failures remain recorded in the snapshot. Searches restricted to an organization’s website cover institutions that lack a verified feed.

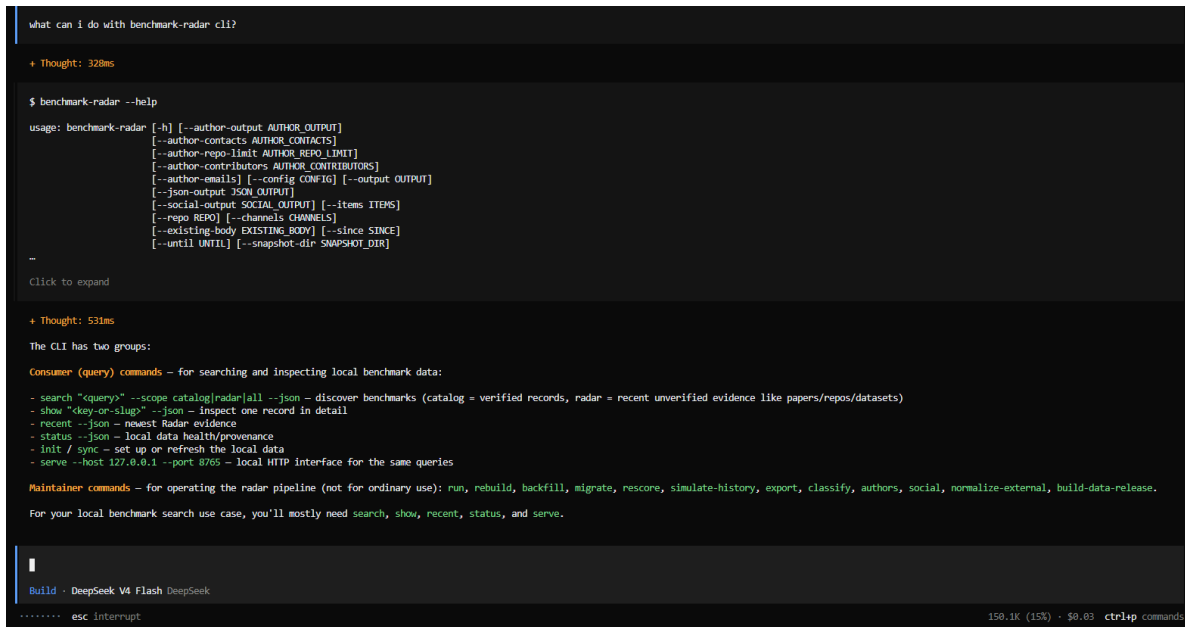
Feeds 1-12	Feeds 13-24
Meituan Engineering	Ai2
OpenAI News	Together AI
Google AI	Sakana AI
Google DeepMind	Qwen
Google Research	Ollama
Apple Machine Learning Research	Stability AI
AWS Machine Learning	Nomic AI
Hugging Face Blog	Replicate
Microsoft Research	NVIDIA Developer
NVIDIA AI Blog	IBM Research
Mistral AI	Databricks
Meta Research	LangChain

Table 10: First-party research and engineering feeds monitored by Benchmark Radar.

D Worked Use Case: Prior-Art Check for a New Evaluation

The contributor wanted to determine whether a proposed evaluation duplicated existing work. They surveyed work published in August 2026 on credit assignment in agentic training, using small Qwen-series models as a requirement for reproducible baseline experiments. The search produced a focused comparison table that the contributor used to assess the proposed design.

The contributor gave the task to a coding agent together with the public setup instructions for Benchmark Radar. The agent installed the command-line client and the Benchmark Radar Skill, a set of instructions for using the client. It then downloaded the local corpus and queried candidate records offline.



```
what can i do with benchmark-radar cli?

+ Thought: 328ms

$ benchmark-radar --help
usage: benchmark-radar [-h] [--author-output AUTHOR_OUTPUT]
                      [--author-contacts AUTHOR_CONTACTS]
                      [--author-repo-limit AUTHOR_REPO_LIMIT]
                      [--author-contributors AUTHOR_CONTRIBUTORS]
                      [--author-emails [CONFIG]] [--output OUTPUT]
                      [--json-output JSON_OUTPUT]
                      [--social-output SOCIAL_OUTPUT] [--items ITEMS]
                      [--repo REPO] [--channels CHANNELS]
                      [--existing-body EXISTING_BODY] [--since SINCE]
                      [--until UNTIL] [--snapshot-dir SNAPSHOT_DIR]

Click to expand

+ Thought: 531ms

The CLI has two groups:

Consumer (query) commands – for searching and inspecting local benchmark data:
- search "query" --scope catalog[radar|all] --json – discover benchmarks (catalog = verified records, radar = recent unverified evidence like papers/repos/datasets)
- show "key-or-slug" --json – inspect one record in detail
- recent --json – newest Radar evidence
- status --json – local data health/provenance
- init / sync – set up or refresh the local data
- serve --host 127.0.0.1 --port 8765 – local HTTP interface for the same queries

Maintainer commands – for operating the radar pipeline (not for ordinary use): run, rebuild, backfill, migrate, rescore, simulate-history, export, classify, authors, social, normalize-external, build-data-release.

For your local benchmark search use case, you'll mostly need search, show, recent, status, and serve.
```

Figure 9: A coding agent installs the Benchmark Radar client and its usage instructions, then runs local queries.

The agent inspected paper, repository, and dataset links to decide which sources to open next. Figures 10 and 11 show two retrieved discovery records: one has a repository link, and the other has both paper and repository links. Neither records a dataset link, leaving dataset availability for a follow-up check.

```
"rank": 18,
"recommended": false,
"score": 18.75,
"slug": null,
"snapshot_date": "2026-09-01",
"source": "Hugging Face Papers",
"source_id": "2608.28476",
"updated_at": "2026-08-28T00:00:00+00:00",
"url": "https://huggingface.co/papers/2608.28476"
},
{
  "categories": [
    "benchmark"
  ],
  "description": "LTABC-KM is a long-tail-aware adaptive artificial bee colony-K-means for unsupervised image clustering. It leverages density-aware coresets and graph-propagation assignment to recover minority tail modes without labels, achieving significant Macro-F1, NMI and ARI improvements on imbalanced benchmarks with favorable quality-computation trade-offs.",
  "has_dataset": false,
  "has_paper": false,
  "has_repo": true,
  "has_size": false,
  "key": "radar:github:Lutra11/LTABC-KM",
  "kind": "radar",
  "languages": [],
  "match": {
    "idf_coverage": 0.4391,
    "matched_fields": [
      "description"
    ],
    "matched_tokens": [
      "assignment"
    ],
    "missing_tokens": [
      "credit",
      "training"
    ],
    "phrase_fields": [],
    "query_coverage": 0.3333,
    "ranking_algorithm": "bm25f_v3",
    "reason": "1 of 3 unique query tokens matched across fields; IDF coverage 0.44",
  }
}
```

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D:\projects\radar-test187.3K (19%) · \$0.04ctrl+p commands

Figure 10: A retrieved discovery record with a repository link for inspecting the implementation. Paper and dataset links are missing from the record.

```
"source": "GitHub",
"source_id": "mdkamranalam/credixa",
"updated_at": null,
"url": "https://github.com/mdkamranalam/credixa"
},
{
  "categories": [
    "benchmark"
  ],
  "description": "Long-horizon agentic tasks require large language models (LLMs) to iteratively retrieve, integrate, and maintain dispersed information across multi-turn interactions, but preserving all interaction histories leads to a continuously growing working context. Recent proactive context management methods allow models to edit their own working context with specialized tools, yet they still face three key limitations: (1) a limited toolset restricted to search, deletion, and summarization, with no support for global planning, long-term memory, and adaptive compression; (2) inefficient exploration that treats context management actions uniformly despite their heterogeneous impacts on final outcomes; and (3) coarse-grained credit assignment that assigns the final trajectory-level reward to all intermediate context editing actions during RL. To bridge these gaps, we introduce ContextPilot, a proactive context management framework for long-horizon agentic reasoning. Our approach systematically augments the toolset with planning, long-term memory, and soft context offloading tools. We further propose an RL method tailored for context management, which uses context and entropy variation to identify critical editing decisions for branch sampling and estimates action-level advantages from all branched trajectories that pass through the corresponding context editing action. Experiments on long-context QA and deep search tasks show that ContextPilot achieves stronger performance with a more compact working context, consistently outperforming existing baselines across various base models and benchmarks. Code is available at https://github.com/Tencent/ContextPilot.",
  "has_dataset": false,
  "has_paper": true,
  "has_repo": true,
  "has_size": false,
  "key": "radar:hugging face papers:2608.28476",
  "kind": "radar",
  "languages": [],
  "match": {
    "idf_coverage": 0.8392,
    "matched_fields": [
      "description"
    ],
    "matched_tokens": [
      "assignment",
      "credit"
    ]
  },
}
```

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Figure 11: A second discovery record provides both paper and repository links. Its dataset link needs a follow-up check.

The agent supplemented local queries with web search to find related work described in different terms. It cross-checked the candidate sets and inspected the source evidence before selecting records for the comparison. Figure 12 shows the word matches and score components available for inspecting one retrieved candidate.


```

"kind": "radar",
"languages": [],
...
Click to expand

$ benchmark-radar search "credit assignment training" --scope radar --json
...
"query_coverage": 0.3333,
"ranking_algorithm": "bm25f_v3",
"reason": "1 of 3 unique query tokens matched across fields; IDF coverage 0.44",
"retrieval_score": 10.824384,
"score_components": {
  "bm25f": 10.824384,
  "name_bonus": 0.0,
  "phrase_bonus": 0.0
},
"modality": null,
"name": "deepvisionaiofficial/Automation-AI-Assignment-Checker",
"openness": null,
"published_at": "2026-08-10T05:56:17+00:00",
"publisher": "",
"rank": 6,
"recommended": false,
"score": 36.32,
"slug": null,
"snapshot_date": "2026-08-10",
"source": "GitHub",
"source_id": "deepvisionaiofficial/Automation-AI-Assignment-Checker",
"updated_at": null,
"url": "https://github.com/deepvisionaiofficial/Automation-AI-Assignment-Checker"
},
{
  "categories": [
    "benchmark",
    "agent"
  ]
}

```

use internal rubrics to compute retrieval score

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D:\projects\radar-test 187.3K (10%) · \$0.04 ctrl+ commands

Figure 12: A candidate record with query-word matches and retrieval-score components for the agent to inspect when judging relevance.

The resulting related-work table appears in Figure 7 in the main text.

Figure 13 shows a comparison table from an earlier benchmark-design effort by one contributor. It covers ten related benchmarks across six design dimensions, with each cell read from a source paper. The contributor reported that assembling it took more effort than any part of that project except producing the benchmark data. Benchmark Radar retrieves candidates for this kind of comparison and exposes the matching words and fields for inspection; the comparison itself requires reading the source evidence.

Bench Name	End-to-End Tasks	Fine-Grained Eval	Researcher-Quality Eval	Data Generation	Multi-Harness Eval	#Tasks
MLE-Bench (Chan et al. 2025)	✗	✓	✗	Transfer&Compose	✓	75
MLGym-Bench (Nathani et al. 2025)	✗	✓	✗	Automatic	✗	13
EXP-Bench (Kon et al. 2025)	✓	✗	✗	Automatic	✓	461
ResearchCodeBench (Hua et al. 2026)	✗	✓	✗	Transfer&Compose	✗	212
MLR-Bench (Chen et al. 2026)	✓	✗	✗	Automatic	✓	201
PaperBench (Starace et al. 2025)	✗	✓	✗	Transfer&Compose	✗	8316
AstaBench (Bragg et al. 2025)	✓	✓	✗	Transfer&Compose	✓	2400+
InnovatorBench (Wu et al. 2025)	✓	✗	✗	Transfer&Compose	✗	20
AIRS-Bench (Lupidi et al. 2026)	✗	✓	✗	Automatic	✓	20
COMPOSITE-Stem (Waters et al. 2026)	✓	✓	✗	Manual	✗	70
ScienceBoard (Sun et al. 2025)	✗	✓	✗	Manual	✗	169

Figure 13: A prior-art comparison table assembled manually for an earlier benchmark-design effort. Each row is one related benchmark and each column one design dimension, all read from the source papers by hand.

This case records one workflow combining Benchmark Radar with web search; it has no measured time or accuracy comparison.

E Reproducibility, Access, and Citation

This revision uses the frozen [Benchmark Radar v0.11.0 release](#), software commit 8f46bbf, with a discovery cutoff of 2026-09-07. The release preserves the inputs already audited for this paper; later data is outside its scope. A fresh detached checkout passed the six required CI steps in order: linting, formatting checks, catalog normalization, classification, release construction, and tests. All 1,314 tests passed. The dated registry inputs remain the August snapshots registered in `data/leaderboard_snapshots.yml`; the September cutoff does not imply that those sources were recrawled on that day.

The data build normalizes the registry and model-report inputs into the shared catalog, then classifies discovery tracks and packages the catalog with checksums. Installed clients validate an update before activating it. The manuscript and its dated analysis files build independently of the software checkout. The project provides a DOI and Citation File Format record for citation [6, 19, 41].

Artifact	Canonical or permanent location
Archived paper (v0.9.0)	https://doi.org/10.5281/zenodo.22167102
Frozen data (v0.11.0)	https://github.com/ktwu01/benchmark-radar/releases/tag/v0.11.0
Source code	https://github.com/ktwu01/benchmark-radar
Dashboard	https://benchmark-radar.org/
Discovery observations (JSON)	https://benchmark-radar.org/data/radar.json
Benchmark catalog	https://benchmark-radar.org/data/benchmark-index.json
RSS	https://benchmark-radar.org/feed.xml
Citation metadata	https://github.com/ktwu01/benchmark-radar/blob/main/CITATION.cff

Table 11: Access and citation artifacts for Benchmark Radar.

The census reads `site/data/benchmark-index.json`, the detail shards under `site/data/benchmarks/`, and the shared document and model registries. Discovery counts come from the rebuilt `site/data/radar.json` and dated snapshots. Model reports and score YAML files are normalization inputs, not a separate population for the paper’s findings.

The paper repository contains `scripts/audit_catalog.py`, which validates IDs and counts, computes the census, and exports `catalog-data.tex` plus `evidence/catalog-audit.json`. The JSON includes every source-record key, measurement state, per-record model and document counts, citation IDs, and SHA-256 hashes of the inputs. The script checks score-observation uniqueness, record-to-shard identity, document-registry coverage, and reconciliation of scored and unscored records. Its `--check` mode compares a fresh calculation against the committed outputs. Figure 8 draws one linked dot per census record in source-key order.

Reproduce from a clean checkout at software commit 8f46bbf using the README’s six-step CI sequence. The software exporter writes `figure-data.tex`; the paper’s `audit_catalog.py` and `audit_examples.py` take the rebuilt software path. The example audit writes `example-data.tex` and `evidence/restored-examples.json`, recording every record’s headroom eligibility, tied maximum-score observations and settings, mention documents and organizations, and the citations behind date gaps. Run each exporter’s `--check` mode, build with `make arxiv`, compile the extracted package, and inspect the PDF. Normal builds use the committed inputs. The v0.9.0 DOI remains a frozen earlier deposit; live dashboard counts require a retrieval date.